

Units of Rotary Motion 1

Essential Pre-Uni Physics F3.1

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P2

How big is 3 rad , when expressed in degrees to the nearest whole number?

Question deck:

STEM SMART Physics 23 - Circular motion

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Essential Pre-Uni Physics F3.8

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P2

A car travels 10 km. One of its wheels has a radius of 30 cm. Calculate the angle the wheel turns as the car travels this distance (answer in radians to 2 significant figures).

Question deck:

STEM SMART Physics 23 - Circular motion

Units of Rotary Motion 3

Essential Pre-Uni Physics F3.3

Subject & topics: Physics | Mechanics | Circular Motion Stage & difficulty: A Level P2

Complete the questions in the table by converting the units.

Time period / s	Frequency / Hz	Angular velocity / rad s^{-1}	Revolutions per minute (rpm)
0.50	(a)	(b)	(c)

Part A

Frequency

a) Frequency?

Part B

Angular velocity

b) Angular velocity?

Part C

Revolutions per minute

c) Revolutions per minute?

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Essential Pre-Uni Physics F3.10

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P2

My washing machine has a spin speed of 1200 rpm, and a drum radius of 20 cm. Calculate how fast clothes go when up against the side of the drum when the machine is spinning. Give your answer to 2 significant figures.

Question deck:

STEM SMART Physics 23 - Circular motion

Centripetal Acceleration 1

Essential Pre-Uni Physics F4.1

Subject & topics: Physics | Mechanics | Circular Motion Stage & difficulty: A Level P2

Complete the questions in the table.

Speed / m s^{-1}	Radius / m	Angular velocity / rad s^{-1}	Centripetal acceleration / m s^{-2}
	0.32	5.2	
2.1	0.070		
	30.0		9.8
		0.20	9.8
60	1200		

Centripetal Acceleration 2

Essential Pre-Uni Physics F4.2

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P2

A car goes round a roundabout at 30.0 mph (13.4 m s^{-1}) on a circular path with a radius of 8.0 m. Calculate the centripetal acceleration.

Question deck:

STEM SMART Physics 23 - Circular motion

Centripetal Acceleration 5

Essential Pre-Uni Physics F4.5

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level C2

A space station with an 8.0 m radius is spun to give the astronauts something which feels like gravity. If the centripetal acceleration is 9.8 m s^{-2} , calculate the speed at which the walls rotate (in m s^{-1}).

Question deck:

STEM SMART Physics 23 - Circular motion

Centripetal Acceleration 6

Essential Pre-Uni Physics F4.6

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level C2

Calculate the centripetal force experienced by a 500 g pair of wet trousers when in the spin cycle of a washing machine with a 20 cm drum radius if it rotates at 1200 rpm. Give your answer to 2 significant figures.

Question deck:

STEM SMART Physics 23 - Circular motion

Cornering on a Smooth Surface

Subject & topics: Physics | Mechanics | Circular Motion**Stage & difficulty:** A Level C2

A car of mass $m = 1000 \text{ kg}$ is driven round a smooth circular track of radius $r = 250 \text{ m}$ and takes a time $T = 30 \text{ s}$ to complete one lap.

At what angle θ must the track be banked to counteract the tendency of the car to slip sideways?

Adapted with permission from UCLES, Higher School Certificate Physics, June 1928, Paper 2, Question 3.

Question deck:

STEM SMART Physics 23 - Circular motion

Geostationary Orbit

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level C2

A satellite is to be placed in a circular orbit around the Earth.

The gravitational force F_A between the satellite and the Earth is in the inward radial direction and its magnitude is given by the equation

$$F_A = \frac{GMm}{R^2}$$

where $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the gravitational constant; $M = 5.97 \times 10^{24} \text{ kg}$ and m are the masses of the Earth and the satellite respectively; and R is the radius of the orbit.

Use the information and data above to calculate the required radius of the orbit if the satellite is in a geostationary orbit (remains above the same point on the equator).

Used with permission from UCLES, A Level Physical Science, June 1989, Paper 2, Question 3.

Question deck:

STEM SMART Physics 23 - Circular motion